

LumenDetect

Analysis of Object Detection Models under Low-Light Environments

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Introduction: The Challenge of Low-Light Object Detection

Object detection: critical for surveillance, autonomous systems.

Traditional detectors: excel in daylight, falter in low light.

Low illumination impacts:

- Feature extraction
- Object localization
- Detection reliability

Real-time systems demand 24/7 reliability. Native capabilities need scrutiny, not just enhancement.



Problem Motivation: Unveiling True Model Robustness

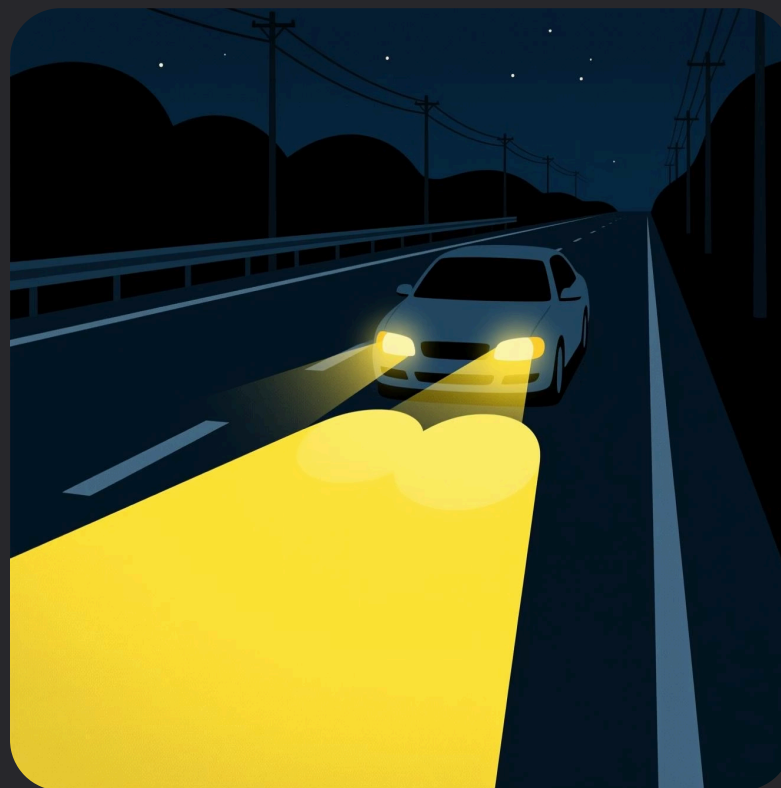
Enhancement methods can introduce artifacts.

True model robustness in dark scenes **remains unknown**.

Research Goals:

- Compare detectors under identical low-light conditions.
- Identify models resilient to illumination drops.
- Benchmark for illumination-aware systems.

Dataset: **ExDark** for consistent low-light evaluation.



ExDark Dataset: A Closer Look

1

7,363 Real-World Images

Authentic low-light scenarios captured in diverse environments.



2

12 Object Categories

Cars, People, Dogs, Buses, and more.

- Class label
- Bounding boxes (l, t, w, h)
- Illumination & environment metadata

3

Diverse Illumination Types

Low, Ambient, Weak, Strong, Window, etc. Structured metadata via **imageclasslist.txt**.

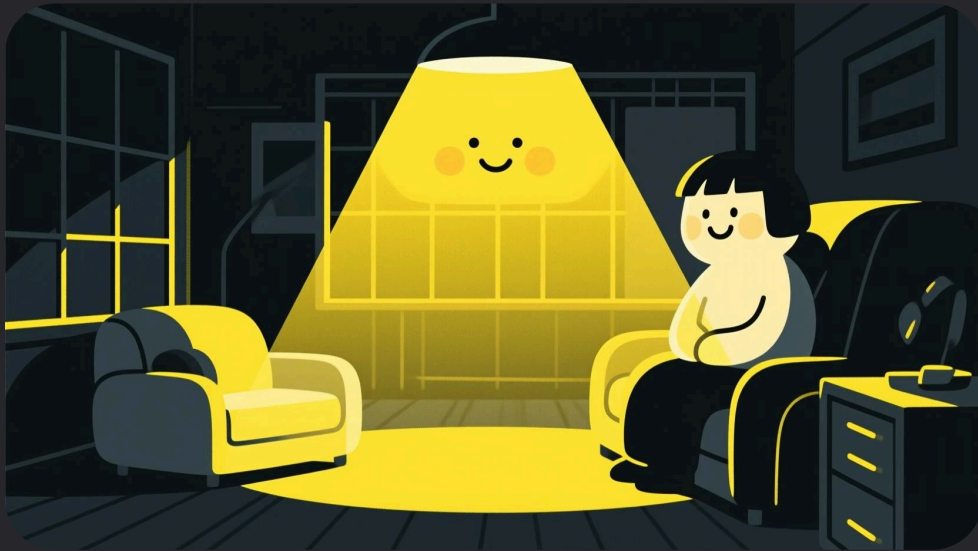


Methodology: Two-Phase Evaluation



Phase 1 – Baseline

Models tested on raw low-light images subset of 2,405 images (across classes: bottle, chair, cup, dog, table). No enhancement applied.



Phase 2 – Enhancement-Based

Applied Contrast Limited Adaptive Histogram Equalization (CLAHE). Evaluated effect on model performance.



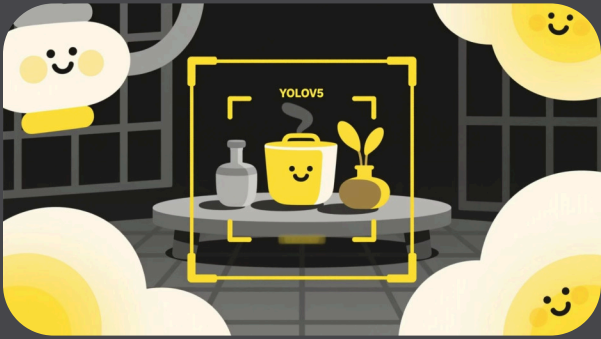
Identical training and testing conditions across all models ensured fair comparison.

Models Under Comparison



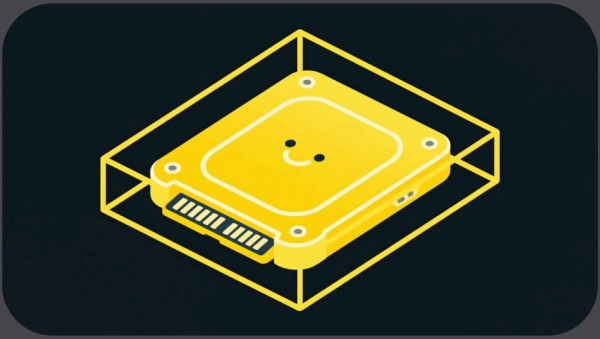
YOLOv5

Single-stage, real-time. Fast inference, strong generalization.



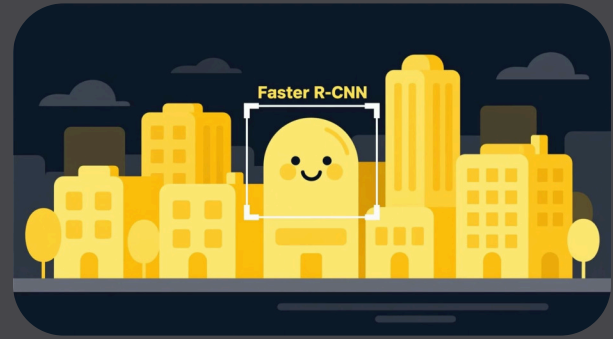
SSD

Single-shot multibox detector. Lightweight, balanced accuracy-speed tradeoff.



Faster R-CNN

Two-stage, region proposal-based. High precision & strong localization.



Enhancement Techniques (Phase 2)

Gamma Correction

- Adjusts luminance curve.
- Improves global brightness and contrast.
- Simple yet effective for overall image clarity.



CLAHE

- Contrast Limited Adaptive Histogram Equalization.
- Enhances local contrast without over-amplifying noise.
- Preserves fine details in varying illumination.



Evaluation Metrics

Precision

Correct positive detections (relevance).

Recall

Correctly detected ground-truth objects (completeness).

F1-score

Harmonic mean of Precision & Recall.

mAP

Mean Average Precision across all classes.
Comprehensive performance indicator.

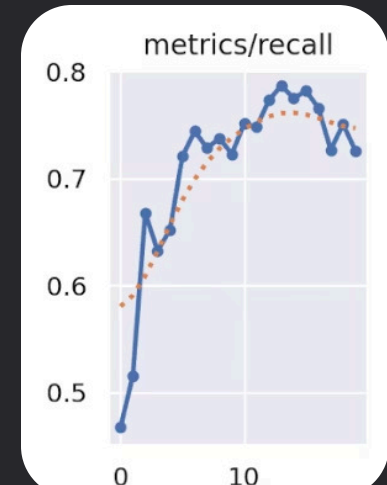
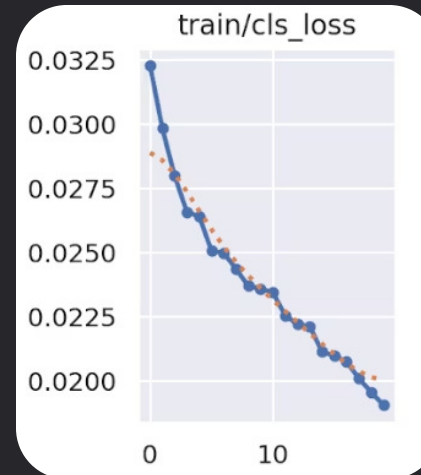
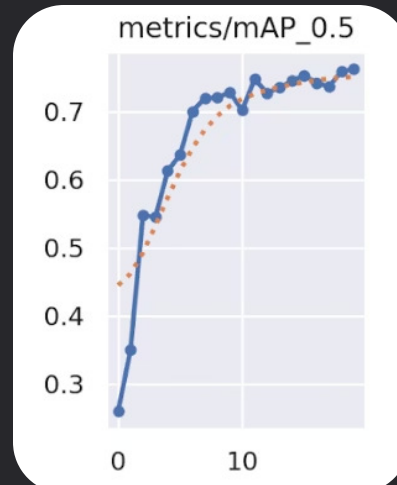
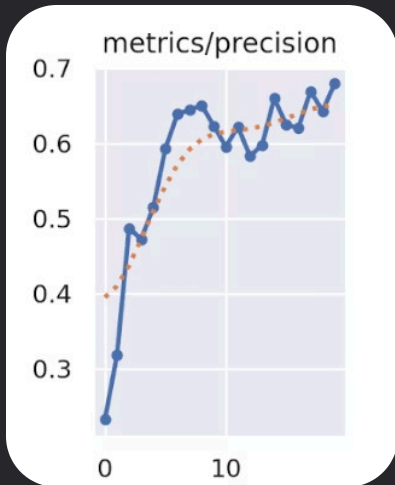
Inference Time

Average processing time per image (efficiency).

Yolov5 model Results & Graphs

Yolov5 model results

- Precision chart
- Recall chart
- mAP 0.5 chart
- Loss chart



Yolov5 model results

Baseline (Raw / Low Light)

Precision (P): 0.685 (68.5%)

Recall (R): 0.766 (76.6%)

Baseline F1 = ~0.72

Inference Time: 531.0 ms/img (CPU)

Enhanced (CLAHE)

Precision (P): 0.748 (74.8%) (↑ +6.3%)

Recall (R): 0.745 (74.5%) (↓ -2.1%)

CLAHE F1 = ~0.75

Inference Time: 522.1 ms/img (CPU)

BASELINE

Class	Images	Instances	P	R	mAP50	mAP50-95: 100% 9/9 [02:29<00:00, 16.62s/it]
all	274	274	0.685	0.766	0.796	0.774
Bottle	274	44	0.78	0.804	0.849	0.803
Chair	274	60	0.598	0.817	0.74	0.728
Cup	274	39	0.566	0.641	0.716	0.706
Dog	274	83	0.882	0.903	0.963	0.928
Table	274	48	0.598	0.667	0.712	0.708

CLAHE

Class	Images	Instances	P	R	mAP50	mAP50-95: 100% 9/9 [02:26<00:00, 16.28s/it]
all	274	274	0.748	0.745	0.817	0.798
Bottle	274	44	0.758	0.773	0.867	0.84
Chair	274	60	0.681	0.748	0.795	0.765
Cup	274	39	0.739	0.667	0.744	0.729
Dog	274	83	0.888	0.892	0.962	0.943
Table	274	48	0.67	0.646	0.718	0.715

SSD Baseline model results

- **Overall Accuracy (mAP@0.5):** Achieved a Mean Average Precision of **64.7%** on unseen test data.
- **High Sensitivity (Recall):** The model demonstrated an exceptionally high Recall of **96.4%** (mAR@100).
- *Viva Explanation:* This means the SSD model almost **never misses** an object in the dark, even if it sometimes generates false positives.
- **Localization Accuracy (mAP Strict):** The Strict mAP (IoU=0.5:0.95) is **60.9%**, indicating that when the model detects an object, the bounding box is drawn very accurately around the target.
- **F1 Score:** The F1 Score (Harmonic mean of Precision and Recall) stands at **0.77**, showing a reasonable balance between finding objects and being correct.

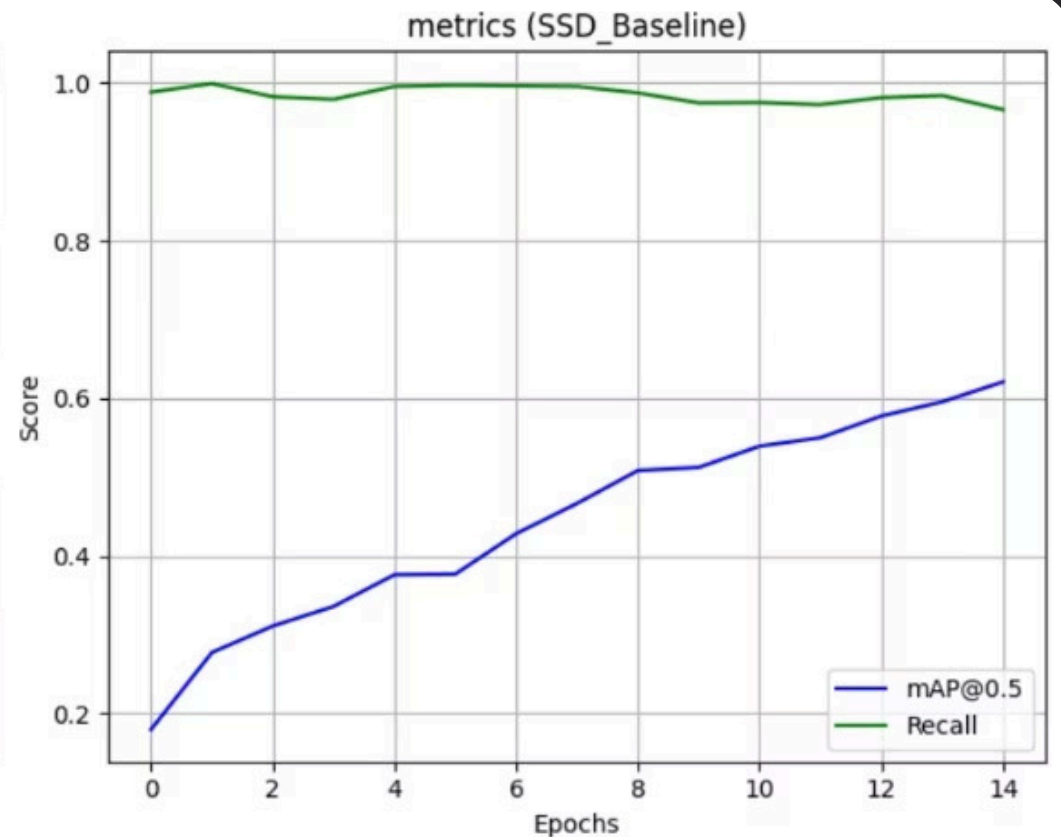
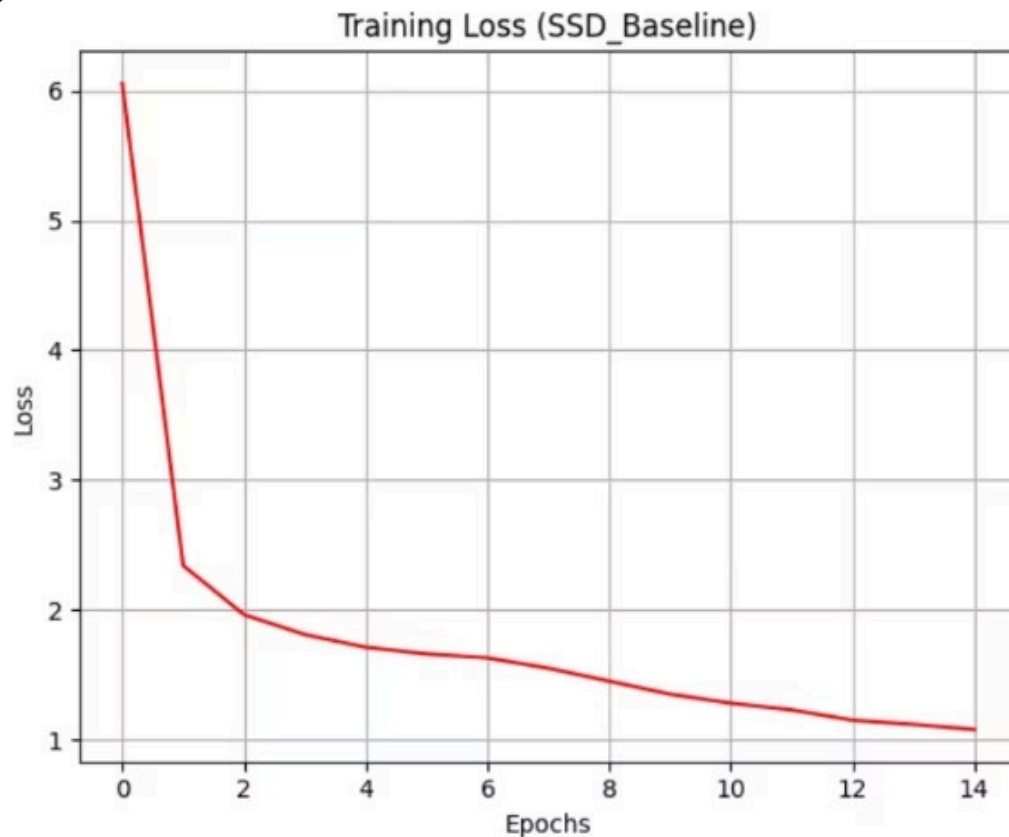
```
=====
📊 DETAILED SSD REPORT
=====
🎯 Precision (mAP@0.5): 0.6474
🔍 Recall (mAR@100):    0.9635
⚖️ F1 Score:            0.7745
=====
```

```
=====
📊 FINAL TEST RESULTS (UNSEEN DATA)
=====
🎯 mAP@0.5: 0.6474
🏆 mAP (Strict): 0.6090
=====
```

Results & Graphs

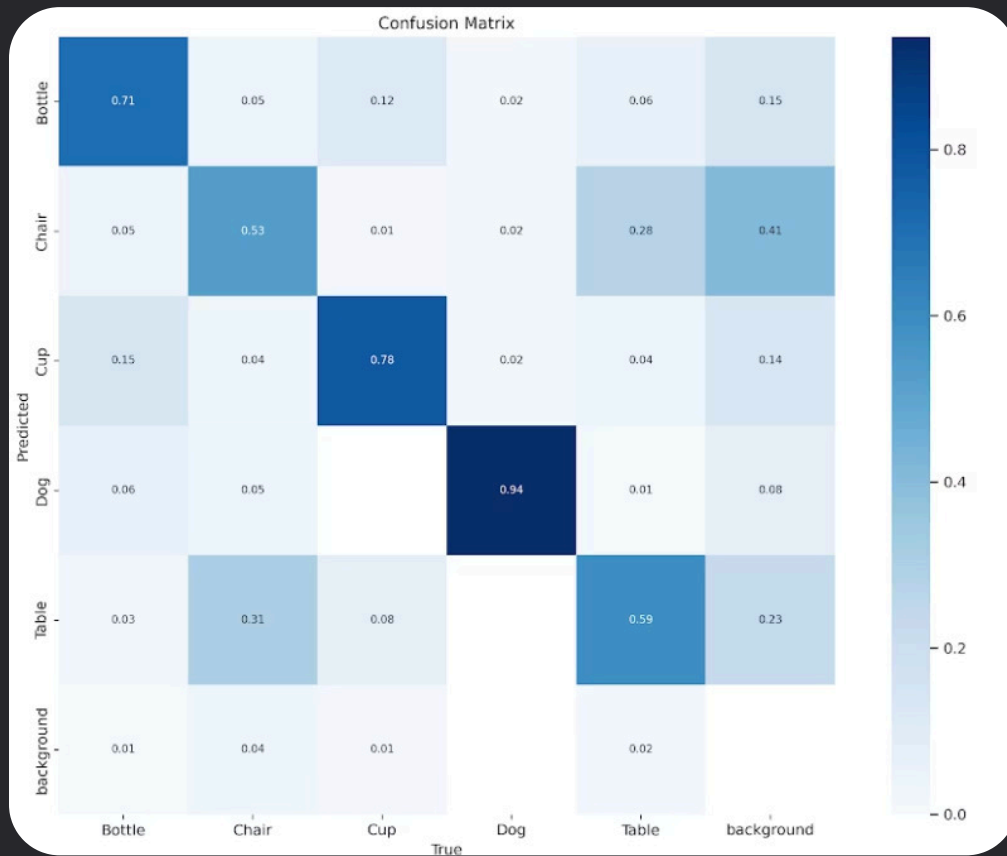
SSD Baseline model results

- Loss Precision chart
- Recall comparison chart
- mAP comparison chart



Results & Graphs

Heatmap comparison of Yolov5.



CLAHE



Baseline

Results & Graphs

- mAP@0.5 improves by +2.4%
- mAP@0.5–0.95 improves by +2.9%
- Recall improves strongly, +6.1%
- Precision improves slightly, +2.4%

Class	Baseline (Dark) mAP50	Enhanced (CLAHE) mAP50	Improvement
ALL (Overall)	76.30%	79.70%	+3.4% 
Table	60.30%	68.60%	+8.3% (Huge!)
Chair	61.30%	64.40%	+3.1%
Dog	93.50%	95.60%	+2.1%
Cup	84.30%	86.20%	+1.9%
Bottle	82.00%	83.90%	+1.9%

improvement calculated from result.csv



Conclusion & Future Scope

This study provides a comparative analysis of key object detection models under challenging low-light conditions.

- Early observations highlight performance disparities between one-stage and two-stage architectures.
- Enhancement pre-processing is expected to significantly influence accuracy.

Thank You

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